

Task 1: Conceptual Understanding

1. What is the difference between "Love" & "low" in NLP?
- ⇒ In NLP, "Love" & "low" are treated as different tokens if the text is case-sensitive. This means the model may consider them as two separate words, even if they have the same meaning. This leads to unnecessary duplication of data. We can change that using normalization.

["Love", "low"] without normalization

["love", "low"] with normalization.

{converting all in lowercase}

2. What happens if stop words are not removed?

⇒ If stop words are not removed they can increase noise in the data set and make the model less efficient. These words don't carry important meanings but appear frequently, which can increase noise and distract the model from learning important patterns.

3. When removing stop words can be harmful?

⇒ In two major cases:-

i) Sentiment Analysis: words like "not" are very important because they determine polarity

ii) Question Answering:- words like "what", "when" and "where" define the intent of the question

4. Difference between stemming & lemmatization

Feature	Stemming	Lemmatization
Method	Rule-based chopping	Dictionary based
Output	Not always real word	Real words
Accuracy	Low	High
Example	"running" ↓ "run" / "runn"	"running" ↓ "run"